

JOSHUA BINU KOCHIYIL

Canberra, ACT, Australia | 0468 883 495 | joshua.kochiyil@gmail.com | [LinkedIn](#) | [GitHub](#)

CAREER OBJECTIVE

Advanced Computing (Honours) graduate with hands-on experience in software engineering and applied machine learning. Designed and delivered full-stack production applications and cloud-based AI solutions across research and startup environments, with a focus on reliability, scalability, and real-world impact. Experienced in developing React-based interfaces, backend services, and operational ML pipelines. Seeking to contribute strong technical execution and ownership while continuing to grow within a collaborative team building impactful software systems.

EDUCATION

Bachelor of Advanced Computing (Honours)
The Australian National University

Canberra, Australia
2021 - 2025

- **Specialisation:** Machine Learning
- **GPA:** 6.31 / 7.00
- Recipient of **ANU Chancellor's International Scholarship**
- **Relevant Coursework:** Software Engineering, Data Structures and Algorithms, Database Management, Systems & Networks Programming, Computer Vision, Natural Language Processing, Bayesian Modelling, Computer Graphics

SKILLS SUMMARY

Web & Frontend: React, Remix, JavaScript/TypeScript, HTML/CSS, Tailwind CSS, Flutter, Figma

Backend & APIs: Python, Flask, FastAPI, Django, RESTful APIs, Java, C/C++

Databases: MySQL, PostgreSQL, Firebase, Supabase

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, SHAP, Explainable AI (XAI), NLP

Cloud & DevOps: AWS, Docker, CI/CD (GitHub Actions), Git, Unix/Linux

Engineering Practices: Test-Driven Development, Technical Documentation, User Story Mapping, Agile SCRUM

Key Strengths:

- **Testing & System Reliability:** Applied structured testing and validation across research and production systems, including unit and integration testing, data validation, and ensuring reliable end-to-end system performance.
- **Collaboration & Agile Delivery:** Worked closely with cross-functional teams and non-technical stakeholders to translate operational requirements into functional digital solutions.
- **Execution & Time Management:** Delivered multiple core system components across parallel sprints in a fast-paced startup environment while consistently meeting project deadlines.

WORK EXPERIENCE

Machine Learning Research Assistant
ANU Research School of Biology

Canberra, Australia
Jan 2025 - Present

- Designed and deployed an **LLM-enhanced SHAP optimisation pipeline** in Python, reducing explanation computation cost by **20%** while maintaining reliable and interpretable outputs.
- Built a **web-based dashboard** that integrates a **conversational AI interface**, increasing **accessibility for non-technical users** by enabling natural-language exploration of machine learning model explanations.
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline with structured **prompt engineering** to ground LLM responses in model-derived explanations, reducing hallucinations and improving response consistency.
- Developed and containerized **production-ready AI microservices** using Docker, ensuring reproducible builds and seamless cloud integration.
- Collaborated with stakeholders to define **success metrics and usability criteria**, ensuring AI outputs aligned with real-world decision-making needs.

Full-Stack Software Developer
Ohna

Canberra, Australia
Feb 2024 - Mar 2025

- Led the design and development of a **real-time order tracking and automation system** (Remix + Supabase), increasing packing throughput by **15%** across **500+ weekly orders**.
- Built and deployed **Python & PostgreSQL automation pipelines** for a high-volume operational database, maintaining data quality and purging stale records to reduce manual overhead and improve query performance.
- Designed **secure relational schemas** within Supabase, implementing **row-level security (RLS)** and role-based access control aligned with cloud security best practices.
- Maintained and optimised a production **Flutter-Firebase e-commerce mobile app** by identifying and resolving critical bugs, improving overall app stability and user experience.
- Secured a **\$30,000 Innovation Connect Grant** from the Canberra Innovation Network by delivering **live technical demonstrations** and presenting the platform's **community and operational value**.
- Led end-to-end delivery within an **Agile Scrum** environment, contributing to **sprint planning** and feature development through structured **Git workflows**.

OTHER EXPERIENCE

Casual Sales Assistant
Shaver Shop

Canberra, Australia
Jul 2025 - Present

- **Customer Engagement & Needs Analysis:** Consistently exceeded sales KPIs through effective customer needs analysis and tailored product recommendations, demonstrating strong communication and active listening.

Casual Retail Team Member
JB Hi-Fi

Canberra, Australia
Jan 2024 - Dec 2024

- **Problem Solving:** Resolved customer complaints and defective product cases through calm, solution-oriented communication.

TECHNICAL PROJECTS

AI Expense Tracker

Tech Stack: Remix, FastAPI, PostgreSQL, SQLAlchemy, Gemini API, Tailwind CSS, Recharts, Docker, GitHub Actions

- Built a full-stack **AI-powered expense tracking platform** with a multi-stage pipeline to extract and structure receipt data using **OCR** and **LLMs**, converting unstructured images into categorised transaction data.
- Developed a **natural language-to-SQL query engine** with strict validation and **user-scoped access** for secure financial queries.
- Developed an interactive analytics dashboard with real-time filtering, trend analysis, and **AI-generated spending insights**, supported by **asynchronous APIs**, **optimised queries**, and **CI/CD with automated testing**.

MADI (Medical Analysis & Diagnostics Intelligence)

Tech Stack: Python, Flask, FastAPI, Scikit-Learn, REST APIs, PyTest, Bootstrap, HTML, CSS

- Architected and deployed a **backend ML inference service** using Flask/FastAPI, exposing **multi-modal diagnostic models** through **REST APIs** for integration with external applications.
- Implemented **unit and integration testing using PyTest** for ML inference endpoints to ensure reliable prediction workflows and data integrity between frontend and backend services.
- Developed a **custom LLM-powered chatbot** that interprets model outputs and generates patient-friendly explanations of diagnostic predictions, improving accessibility of complex medical insights.

ACHIEVEMENTS

- **McCusker Prize in Humanitarian Engineering (2024):** Awarded for outstanding real-world impact of project *Ohna* in tackling food insecurity through technology, demonstrating engineering excellence with humanitarian value.
- **ANUEC Hackathon Finalist (2024):** Advanced to the final round in the ANU Entrepreneurship Club (ANUEC) 24-hour hackathon with project *MADI*. Demonstrated the project's technical depth and social impact to a panel of industry experts, highlighting innovation in healthcare accessibility and technical execution.